RAJALAKSHMI ENGINEERING COLLEGE

B.E. Computer Science and Engineering (Cyber Security)

CR23532 – CYBER AND DIGITAL FORENSICS

Case Study Assignments – II

Field	Details
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Class / programme	B.E. Computer Science and Engineering (Cyber Security)
Course	CR23532 – Cyber and Digital Forensics
Submission deadline	16 October 2026, before 4:00 PM

Unique scenarios selected

Assignment	Case no.	Unique scenario selected
2(1)	GEC-CDF-2026-A1	Out-of-hours copying of the Nov/Dec 2026 question-paper bank from Linux server EXAM-SRV02 via account r.mohan
2(2)	GEC-CDF-2026-A2	Hidden printspool_helper.exe and encrypted archives on the shared Library desktop LIB-DESK-04 (intern2 account)
2(3)	GEC-CDF-2026-A3	Seized laptop and USB drive of an outsourced network technician; BitLocker-protected SSD found powered on
2(4)	GEC-CDF-2026-A4	Scholarship_Disbursal_Oct2026.xlsx altered and deleted on ACC-PC01; versions recovered from two Volume Shadow Copies
2(5)	GEC-CDF-2026-A5	Unauthorized faculty-level access to GEC-LMS and grade changes; account arvind.p under suspicion

Contents

Part	Title	Questions
Assignment 2(1)	Suspected Unauthorized Access to a College Server	Q1–Q7
Assignment 2(2)	Linking a Suspicious File to a User Using File-System Evidence	Q1–Q7
Assignment 2(3)	Forensic Imaging and Safekeeping of a Suspect Storage Device	Q1–Q8
Assignment 2(4)	Recovery of Deleted Evidence from Volume Shadow Copies	Q1–Q8
Assignment 2(5)	Multi-Source Investigation of a Suspected Cyber Incident	Q1–Q8
Appendices	Glossary, standards and laws referred to, overall conclusion	A, B

Declaration and general notes

- All scenarios, names, institutions, times, hash values, serial numbers and IP addresses in this report are **hypothetical** and were chosen for illustration. Each assignment states its assumptions at the start.
- Where a tool (ILookIX, IXImager, ASB container, Event/Cloud/Lead Analysis) is mentioned, the report explains the **intended forensic workflow**. No operation is claimed to have been performed and no screenshots are included.
- Every assignment follows the five-phase digital forensic process: Identification, Collection, Examination, Analysis and Presentation, and keeps observed facts separate from inference.
- Each assignment includes the deliverables requested: flowchart or workflow, evidence/classification tables, timeline or correlation matrix, and a conclusion with limitations.

Assignment 2(1): Suspected Unauthorized Access to a College Server

Unique scenario selected: *Out-of-hours copying of the Nov/Dec 2026 end-semester question-paper bank from the Linux file server EXAM-SRV02 through the staff account r.mohan.*

Assumptions and case-file identifiers (hypothetical)

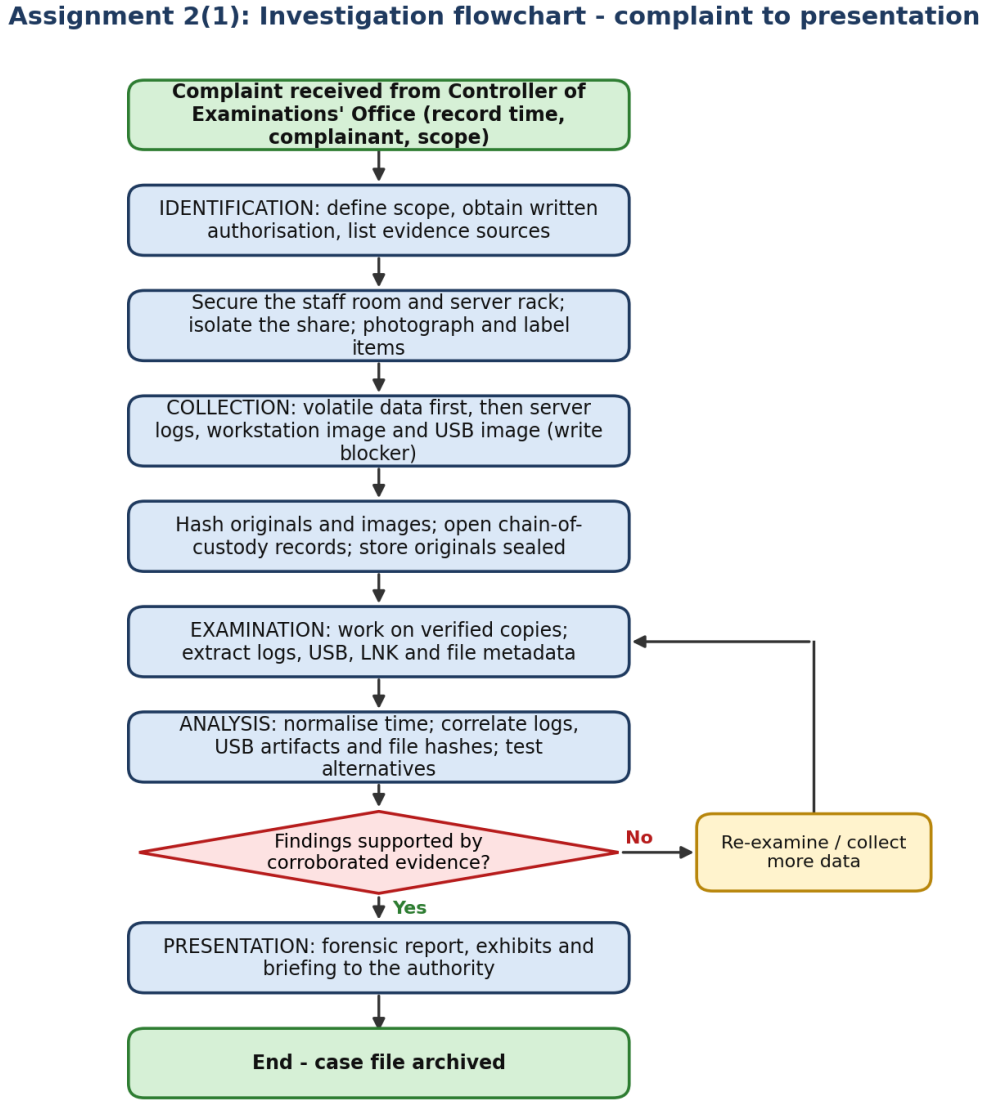
Field	Value
Institution	Greenfield Engineering College (GEC), Chennai (hypothetical) – Dept. of CSE (Cyber Security)
Case number	GEC-CDF-2026-A1
Server	EXAM-SRV02 – rack server, Ubuntu Server 22.04 LTS, IP 172.16.8.21, Server Room, Block C. Samba share **[qpbank]** (/srv/qpbank) with **vfs_full_audit** logging to /var/log/samba/audit.log; SSH enabled; time kept by chrony (NTP)
Suspect workstation	STAFFROOM-WS15 – Windows 10 Pro 22H2, IP 172.16.8.115, Staff Room D-207, Block D
Account under suspicion	r.mohan (Lab Instructor, Dept. of ECE; has legitimate read access to the share only during working hours)
USB device	Transcend JetFlash 790, 16 GB (serial recorded from USBSTOR)
Files of interest	18 files under /srv/qpbank/NovDec2026/ (for example CS3451_QP_SetA.pdf, EC3452_QP_SetB.docx)
Suspected window	Sunday 11 Oct 2026, 00:30 to 01:15 IST (outside normal working hours)
Reported by	Dr. S. Priya, Controller of Examinations' Office, Monday 12 Oct 2026, 09:15 IST, after a weekly audit review
Authorisation	Principal's written order Ref. GEC/PRIN/2026/112 (scope: server, STAFFROOM-WS15, USB, related logs)

The complaint is treated as an investigative lead, not as proof. All names, times and identifiers are invented for illustration; no tool is claimed to have been run.

Q1. How the team moves through the five phases

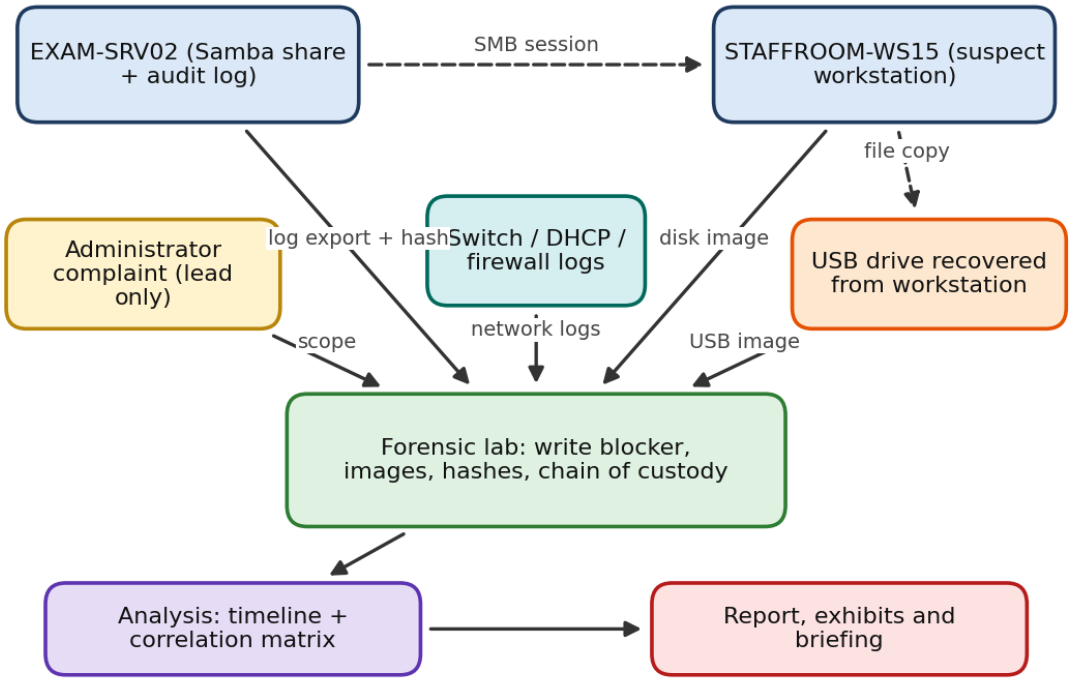
Phase	Activities in this case	Output
Identification	Log the complaint (who, when, what); define scope (server, workstation, USB, logs, 18 files); confirm written authority; list evidence sources and persons with knowledge; identify volatile data (running sessions, ARP/DHCP state, RAM of the workstation if powered on) and log-rotation deadlines.	Case file, evidence inventory, collection plan
Collection	Secure staff room and server rack; photograph and label items; capture volatile data first; export Samba audit, auth.log, wtmp and DHCP/switch logs with hashes; image the workstation drive and the USB drive through a write blocker; seal, tag and start chain of custody.	Forensic images, log exports, hash list, custody forms
Examination	Verify hashes; work only on copies; extract accounts, USB history (USBSTOR, MountedDevices, setupapi log), LNK files and jump lists, Event Log entries; filter audit.log for the share and time window; recover deleted items.	Artifact set, filtered log sets, notes
Analysis	Normalise all times to UTC (IST noted); correlate logon, file-access, USB and file-metadata events; compare file hashes on the USB with server originals; test alternatives (shared password, remote access, clock error).	Timeline, correlation matrix, findings with confidence
Presentation	Write the report with methods, tool versions, hashes, limitations and exhibits; give a plain-language summary to the college authority; retain the case file for disciplinary or legal use.	Final report and briefing

Q2. Forensic investigation flowchart



Flowchart 1: From the initial complaint to presentation of findings

Figure 1: Evidence sources and flow into the forensic laboratory



Q3. Potential digital evidence and its investigative value

#	Evidence item	What it can show	Value	Class
1	Samba audit log and auth.log on EXAM-SRV02	Which account opened which files, from which IP, at what time; successful and failed logins	High	Primary
2	Disk image of STAFFROOM-WS15	User profiles, recent files, mapped share, deleted items, browser and application traces	High	Corroborating
3	USB drive (Transcend JetFlash 790)	Copied files, file-system timestamps, volume serial; hash match with server originals	High	Primary
4	The 18 paper files on server and USB	Hash comparison, created/modified/accessed times, embedded metadata	High	Primary
5	Workstation registry and Event Logs (USBSTOR, MountedDevices, 4624/4647, Kernel-PnP)	First/last connection time of the device, volume serial, active session	Medium-High	Corroborating
6	DHCP, switch and firewall logs	IP-to-MAC mapping, source of the SMB session	Medium	Corroborating
7	Staff-room door-access and CCTV records	Physical presence and opportunity	Medium	Contextual
8	Administrator's complaint, permission and rota records	Reported time, normal access rights, who was on duty	Low-Medium	Contextual

Classification: *primary* evidence directly records the access or holds the copied data (items 1, 3, 4); *corroborating* evidence ties the activity to a machine, session or time (items 2, 5, 6); *contextual* evidence explains opportunity and authorisation (items 7, 8). A complaint alone is never proof of what happened.

Q4. Precautions during collection and preservation

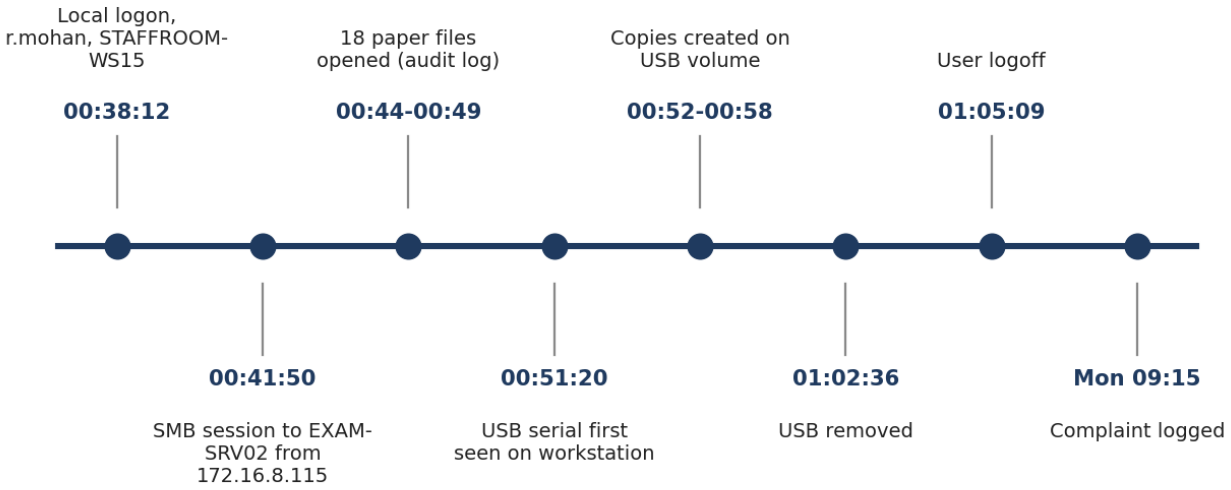
- **Authority first:** obtain written approval and define systems, accounts, date range and data categories before touching anything.
- **Order of volatility:** capture RAM, active sessions and network connections of the workstation before power-off if it is running; record the tool, time and reason. Export server logs immediately, before rotation.
- **No contamination:** never browse the suspect machine normally; use a hardware write blocker for the workstation drive and USB; never copy files onto the original media.
- **Document:** photograph screen, ports, cables and labels; record make, model, serial numbers, date, location, collector and the system-clock offset against a trusted time source.

- **Hash and verify:** compute SHA-256 (and MD5 if required by lab policy) for every image and log export at acquisition and again before analysis.
- **Seal and store:** tamper-evident anti-static bags, unique evidence IDs, locked storage with restricted access.
- **Chain of custody:** record each person, time, purpose and location for every transfer, with no gaps and with signatures.
- **Work on copies:** keep originals untouched so that any result can be reproduced from the verified image.

Q5. Establishing a timeline of the suspected unauthorized access

Investigators extract timestamps from the server audit and authentication logs, the workstation Event Log, USB registry artifacts, LNK/jump-list entries and file-system metadata. For each entry they record the original value, source, time zone, known clock offset and the normalised time (UTC, with IST shown). Events are then joined by account, IP/MAC address, file path, volume serial and file hash. The sequence below is **hypothetical**; each line must be backed by an actual artifact in a real case.

Figure 2: Illustrative event timeline (IST, hypothetical)



Time (IST)	Source	Event	Caveat
11 Oct 00:38:12	Workstation Security log, Event 4624 (type 2)	Interactive logon, r.mohan	Confirm clock offset; shows account, not person
00:41:50	EXAM-SRV02 Samba log / auth	SMB session established from 172.16.8.115	Match IP to MAC via DHCP
00:44:03-00:49:31	audit.log (full_audit: open, pread)	18 files in NovDec2026 opened and read	Check audit configuration was active
00:51:20	Registry USBSTOR / setupapi.dev.log	Device first seen on workstation	Connection does not prove copying
00:52:10-00:58:44	USB file system ($MFT created times)	18 files created on volume	Compare hashes with server originals
01:02:36	Kernel-PnP event / registry last-removal time	USB removed	Time source reliability
01:05:09	Security log, Event 4647	User-initiated logoff	Session end
12 Oct 09:15	Complaint register	Report logged, case opened	Discovery, not event time

Gaps (for example a missing log hour) are stated openly and are not filled by assumption.

Q6. Final forensic report and presentation to a non-technical authority

1 Case title, case ID, requesting authority, examiner and dates.

2 Executive summary in plain language (half a page).

3 Allegation, scope, questions to answer, authorisation and assumptions.
4 Evidence inventory with identifiers, condition, hashes and chain-of-custody references.
5 Methods: tools and versions, acquisition details, time-zone handling, validation steps.
6 Findings by evidence source, with exhibits and the timeline; each finding with a confidence level.
7 Correlation and interpretation, including alternative explanations.
8 Limitations, missing evidence and unresolved questions.
9 Conclusion proportionate to the evidence, separating observed facts from inference; recommendations (access control, log retention, monitoring).
10 Appendices: hash manifest, log excerpts, screenshots, timeline, custody records, examiner declaration.

Presenting to the Principal or Disciplinary Committee: use short sentences, explain each technical term once, show a one-page timeline graphic, separate *what the evidence shows* from *what it suggests*, state clearly what it cannot prove (for example, that the account was used, but not by whom), and avoid accusing any named person.

Q7. Legal and ethical considerations in a college environment

- **Authorisation and policy:** the IT acceptable-use policy and written management approval define what may be examined.
- **Privacy and minimisation:** collect only data relevant to the authorised scope; do not read personal mail or unrelated student/staff data.
- **Legal framework:** the Information Technology Act, 2000 (for example Sections 43 and 66 on unauthorised access and computer-related offences) and the Bharatiya Sakshya Adhiniyam, 2023 (electronic records and the certificate requirement) guide admissibility; escalate to police or legal counsel if an offence is likely.
- **Fairness:** presume innocence, avoid bias, preserve exculpatory as well as inculpatory material.
- **Confidentiality:** exam data and the report are shared only with authorised persons.

Conclusion for Assignment 2(1)

A defensible conclusion needs independently corroborated evidence, verified acquisition, a transparent timeline and honest limits. In this hypothetical case the logs, USB artifacts and matching hashes would show that the account r.mohan, the workstation STAFFROOM-WS15 and the USB device were involved; identifying the individual operating the account would need further evidence such as door-access records or an interview.

Assignment 2(2): Linking a Suspicious File to a User Using File-System Evidence

Unique scenario selected: *A hidden printspool_helper.exe and encrypted archives found on the shared Library Digital Section desktop LIB-DESK-04, possibly linked to the intern2 account.*

Assumptions and case-file identifiers (hypothetical)

Field	Value
Case number	GEC-CDF-2026-A2
Computer	LIB-DESK-04 – desktop PC, Library Digital Section, Block A; Windows 10 Pro 22H2; NTFS system volume; one 500 GB internal HDD
Local accounts	libadmin (Librarian), intern1 (project intern), intern2 (project intern), visitor (should be disabled by policy but found enabled)
Suspicious executable	printspool_helper.exe, about 1.1 MB, SHA-256 5d3c...a9f0 (hypothetical, shortened)
Path found	C:\Users\intern2\AppData\Roaming\PrintHelper\printspool_helper.exe
Download source	hxxps://files-sharehub[.]example/tools/printspool_helper.exe (defanged; treated as an untrusted external host)
Hidden files	pdx1.dat, pdx2.dat, pdx3.dat with hidden + system attributes in the same folder
Encrypted archives	fines_ledger_2026.7z and member_export.7z (7-Zip, AES-256) in the same folder
Reported by	Librarian, after an antivirus alert on 09 Oct 2026
Authorisation / image	Principal's order Ref. GEC/PRIN/2026/118; verified disk image with SHA-256 recorded in the acquisition log

ILookIX steps below describe the **intended workflow** from the syllabus. No operation is claimed to have been performed and no screenshots are included.

Q1. Hardware and software components to document before analysis

Component	Items to record	Why it matters
Computer hardware	Make, model, serial number, CPU, RAM, attached peripherals, BIOS/UEFI version, clock difference from a trusted source	Identifies the system; guides acquisition; supports timestamp correction
Storage devices	Drive model, serial, capacity, interface, partition scheme (MBR/GPT), health indicators	Shows where evidence can be; chooses imaging method
Operating system	Name, edition, build, install date, hostname, time-zone setting, update level	Gives context for artifact locations and time conversion
File systems	NTFS/FAT/exFAT, volume labels and serials, allocated and unallocated areas, shadow copies, encryption	Decides which metadata and recovery methods exist
Accounts and permissions	Account names, SIDs, groups, profile folders, last logon, password policy, guest/shared accounts	Needed for attribution of activity to an account
Applications	Browsers, 7-Zip/WinRAR, remote-access tools, security software, startup items, services	Explains artifacts and possible benign sources
Network / removable devices	Adapters, MAC addresses, mapped drives, USB history	Shows other paths for file transfer
Forensic tooling	ILookIX version, hash algorithm, examiner, date and time	Makes the work repeatable

Q2. How storage, operating systems, file systems and metadata help locate evidence

- **Storage devices** keep allocated files, deleted entries, slack and unallocated space, and hidden partitions; the serial number ties a drive to a machine.
- **Operating system** artifacts (user profiles, NTUSER.DAT, AppData, Recent items, LNK files, jump lists, event logs, registry, Prefetch) show who ran or opened what.
- **File system (NTFS)**: the Master File Table stores names, parent path, size, attributes and timestamps; the journal and slack space may hold traces of deleted or renamed files.
- **Metadata**: MACB timestamps, owner SID, attributes (hidden, system), alternate data streams and embedded document properties place a file in time and link it to an account.

- Used together, the examiner starts from the suspicious file, finds its path and owner, then follows related artifacts (download record, execution trace, session records).

Q3. Effect of hidden and encrypted files; evidence-handling considerations

Hidden files carry the hidden/system attribute, sit in unusual folders or use misleading names. They slow a manual search but are normally visible to forensic tools that read the file system directly. Encrypted files may be unreadable without a key, which blocks content analysis, yet name, size, timestamps and container headers usually remain available. Hidden or encrypted status alone is not proof of wrongdoing.

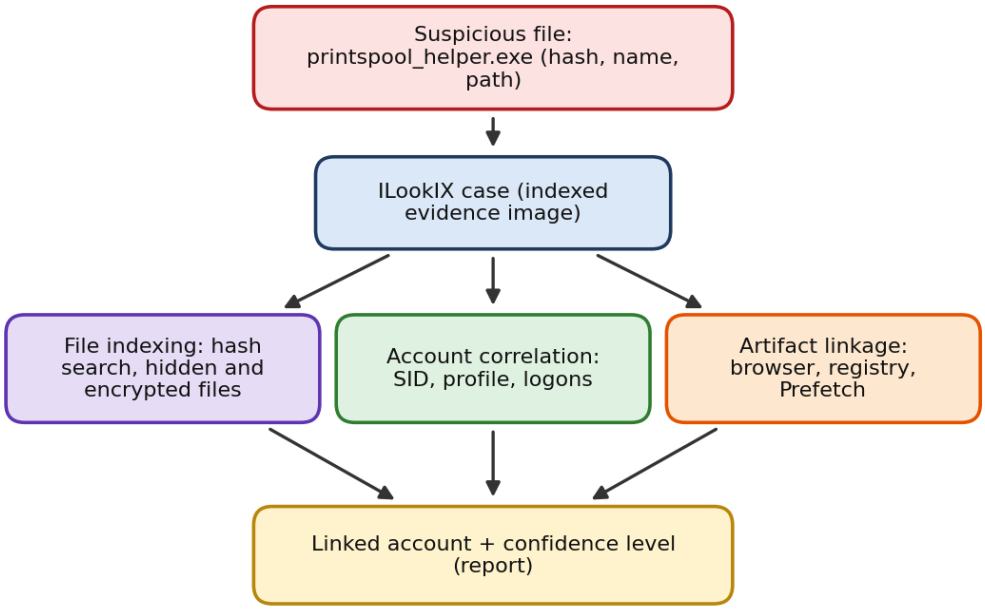
- Work only from the verified image; never alter attributes or attempt decryption on the original.
- Record path, attributes, size, timestamps and hash of every hidden or encrypted item, and how it was found, before any decryption attempt.
- Look for keys or passwords only by lawful, documented means within the authorisation (running processes, key files, credential stores); do not exceed the legal remit.
- If decryption fails, report that the content is unavailable and rely on metadata and surrounding artifacts; do not guess the content.
- Note whether the encryption tool (7-Zip) is installed on the system and who ran it, since this is a contextual clue.

Q4. Evidence-linkage table

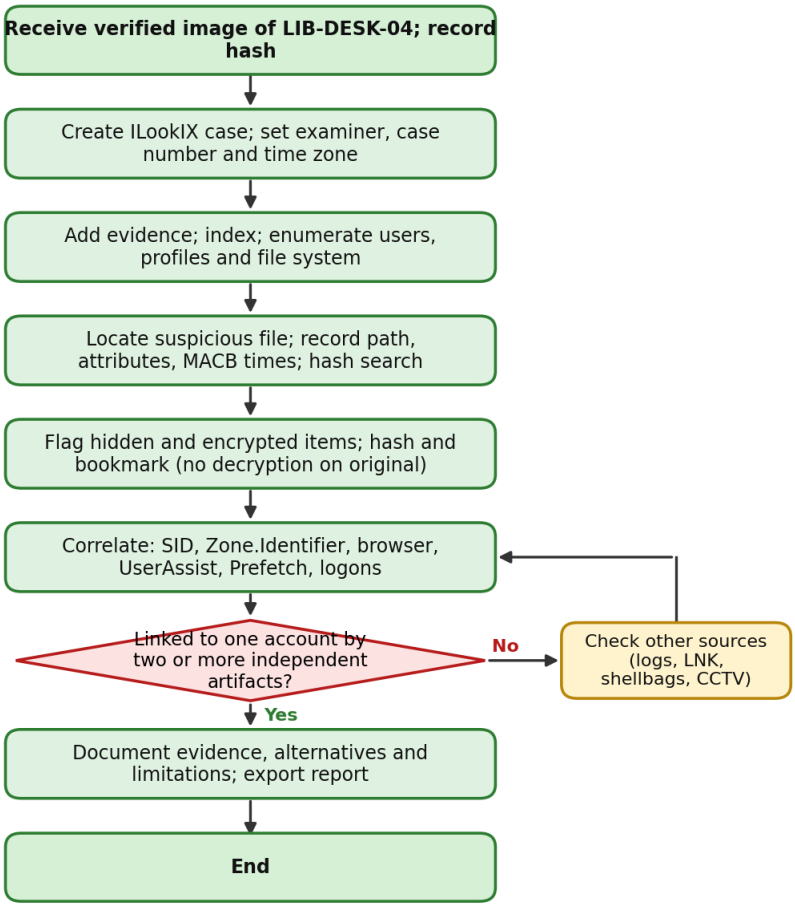
Artifact	Example finding (hypothetical)	How it points to a user	Strength
File name and path	...\Users\intern2\AppData\Roaming\PrintHelper\printspool_helper.exe	Located inside the intern2 profile	Moderate
Owner SID	Owner SID ends in the RID of intern2	Ownership set at creation by that account	Moderate-High
Timestamps	Created 07 Oct 2026 19:42:18; modified 19:42:18; accessed 09 Oct 10:05	Creation falls inside an intern2 session	Moderate
Logon records	Event 4624 intern2 at 19:30:05; logoff 20:11:47	Session covers the creation time	High if only one session was active
Browser download	Edge download entry at 19:41:55 under the intern2 profile	Shows how and by which profile the file arrived	High
Zone.Identifier stream	ZoneId=3 with the host URL on the executable	Confirms an internet origin (mark of the web)	Supporting
Execution traces	UserAssist (intern2 NTUSER.DAT) run count 2; Prefetch PRINTSPOOL_HELPER.EXE-xxxx.pf with two run times	Shows execution and the profile that ran it (UserAssist is per user)	High
Hidden files	pdx1-3.dat created 19:42-19:43, hidden + system	Same folder and time cluster as the executable	Contextual
Encrypted archives	Archives created 19:47 and 19:49, after the download	Timing supports a sequence of activity	Contextual

Q5. ILookIX workflow for importing and analysing the evidence

Figure 3: Linking a suspicious file to a user account with ILookIX



ILookIX analysis workflow (intended procedure)



Flowchart 2: ILookIX analysis workflow (intended procedure)

1 **Prepare:** verify and record the image hash; create a new case; enter case number, examiner and time-zone settings.

2. **Import:** add the forensic image as an evidence item; the tool reads partitions and file system without changing the source.
3. **Process and index:** index names, content and metadata so they are searchable; carve deleted files if needed.
4. **Explore:** browse profiles, show hidden and deleted files, sort by timestamps, examine registry and browser artifacts.
5. **Search and filter:** search the file name, hash and related strings; filter by account and date window; look for the same hash elsewhere.
6. **Bookmark and note:** tag relevant items, write why each matters, export the bookmarked set with hash values.
7. **Report:** generate the case report, add examiner notes, alternatives and limitations.

ILookIX step	Evidence obtained	Interpretation
Add evidence	Partition and volume structure	Confirms what the drive contained
Index / process	Searchable names, text, metadata	Speeds up locating relevant items
File-system browsing	Paths, attributes, MACB times	Shows where and when files exist
Registry and user analysis	Accounts, UserAssist, recent-file records	Links activity to a profile
Search and filter	Items matching name, hash or date	Removes noise and shows patterns
Bookmark and report	Exported items with hashes	Reviewable, repeatable evidence

Q6. Direct versus supporting (contextual) artifacts

Type	Meaning	Examples in this case
Direct (user-linking)	Records a specific action in a relatively explicit way, tied to one account through an identifier	UserAssist entry in intern2's NTUSER.DAT; download record in intern2's Edge profile; owner SID of the file
Supporting / contextual	Adds background or timing but could fit more than one person	File sitting in a shared machine; timestamps alone; hidden files; encrypted archives; account enabled by mistake

Even a direct artifact identifies an *account or process*, not necessarily the human who controlled it, so it is strengthened by independent evidence such as door-access or CCTV records, or a password known to one person only.

Q7. Limitations and alternative explanations

1. **Shared or compromised credentials:** another person, malware or a remote session could have used the intern2 account.
2. **Timestamp uncertainty:** wrong clocks, time-zone settings, or copy operations can shift apparent times.
3. **File origin:** the executable might have been placed by another administrator, a script or an automatic update.
4. **Shared use:** an unattended logged-in session could have been used by a different person; the visitor account was enabled.
5. **Missing data:** cleared logs, rotated records or an undecryptable archive leave gaps that must be reported.
6. **Background processes:** antivirus scans, indexing or backup agents may touch files and alter access times.

Conclusion for Assignment 2(2)

A defensible linkage names the exact account, device, file and artifacts relied on, states the confidence level, and lists alternatives that were considered. In this hypothetical case the profile path, SID, download record, UserAssist/Prefetch and the logon session all point to **the intern2 account**; this supports, but does not by itself prove, that a particular person acted. No single artifact is conclusive, and ILookIX results must be validated by the examiner.

Assignment 2(3): Forensic Imaging and Safekeeping of a Suspect Storage Device

Unique scenario selected: *Seizure of the laptop and USB drive of an outsourced network technician suspected of copying the alumni database, with a BitLocker-protected SSD found powered on.*

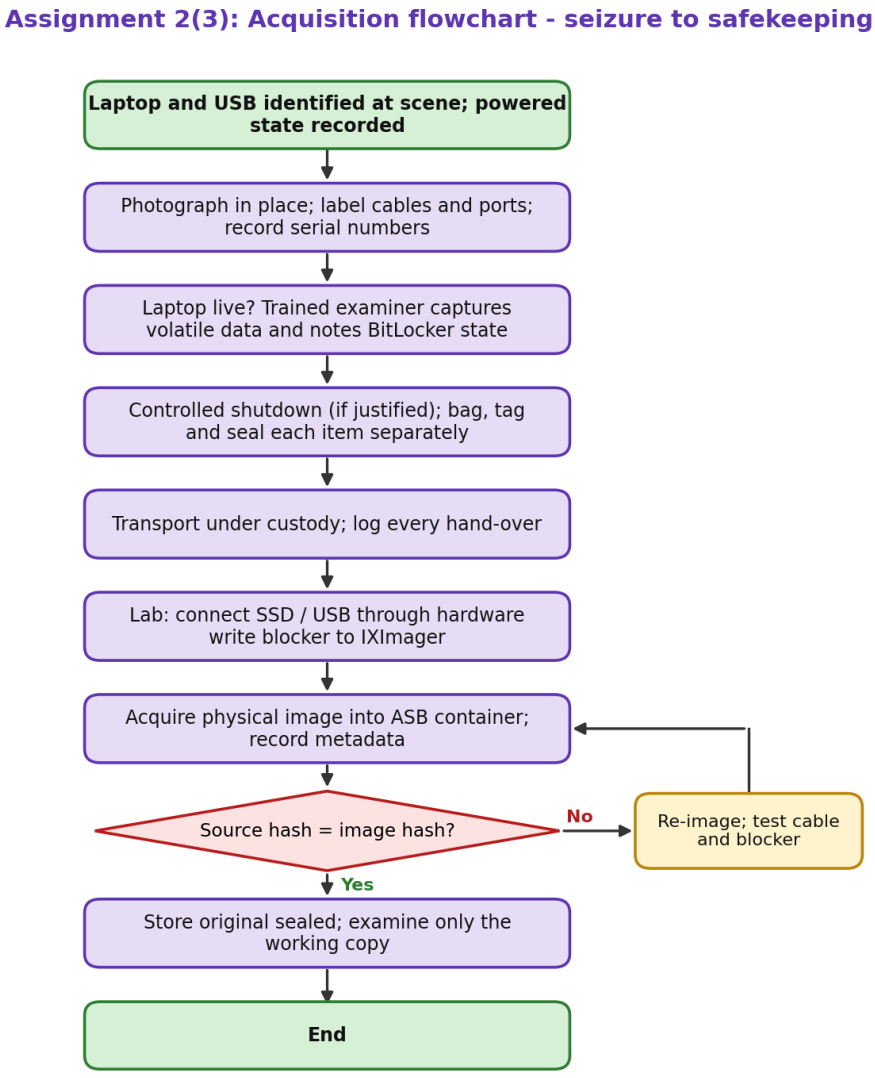
Assumptions and case-file identifiers (hypothetical)

Field	Value
Case number	GEC-CDF-2026-A3
Subject	Mr. Karthik S., outsourced Network Technician (vendor staff), Network Operations Room, Block A
Item L-01	HP ProBook 440 G8 laptop, serial and asset tag recorded from label; 500 GB SATA SSD (serial recorded); Windows 11 Pro with BitLocker
Item L-02	Transcend JetFlash 790, 128 GB USB drive, attached to the laptop at seizure
State at seizure	Laptop powered on and unlocked, VPN client connected, a file-transfer window open; USB drive plugged in
Seizure	13 Oct 2026, 11:20 IST, by Mr. T. Ganesan (Deputy Registrar, authorised collector) with the Digital Forensics Cell
Roles	Evidence custodian: Ms. Anitha V.; Examiner: R. Arulkumar (Forensic Examiner); Analyst: M. Farhan
Tools	IXImager (version recorded from the tool at acquisition); ASB container; hardware write blocker (model and serial recorded)
Authorisation	Principal's order Ref. GEC/PRIN/2026/121; vendor management informed; police Cyber Cell informed because the subject is a non-employee
Destination	Forensically wiped 2 TB drive with enough free space for both images

IXImager and ASB steps describe the **intended workflow**; exact menu options and ASB features must be confirmed against the installed product documentation. Nothing here claims an operation was actually performed.

Q1. Procedure for collecting and transporting the laptop and USB device

1. Secure the area, keep people away from the devices, and note everyone present.
2. Photograph the laptop and USB drive in place (screen, cables, ports, labels) and sketch the layout; record date, time, location and collector.
3. Do not explore files or close windows. Because the laptop is on and protected by BitLocker, a trained examiner follows the approved live-response procedure: photograph the screen, record running programs, then capture RAM and network/VPN state if authorised, noting the tool and time.
4. Record the encryption state and obtain the BitLocker recovery key lawfully (for example from the college's key escrow) so that the volume can be read after shutdown.
5. Cut network access (disable Wi-Fi, unplug cable) to prevent remote wiping; record the decision and reason.
6. Document the USB connection, then remove it only if safe; label it separately. Shut the laptop down per policy and record the method and time.
7. Place each item in its own tamper-evident anti-static bag; write case number, item number, time and signature on the seal; keep chargers and accessories labelled.
8. Transport in a padded container away from heat, magnets and moisture, in the continuous custody of the collector; log each hand-over.
9. At the lab, check seals and condition, log receipt, and store in a locked evidence locker with restricted access.



Flowchart 3: Acquisition flowchart from seizure to safekeeping of the original

Q2. Purpose of physical acquisition and why analysis uses an image

A **physical acquisition** copies the accessible addressable contents of the device sector by sector: allocated files, deleted data, slack space, unallocated space, partition tables and, where supported, hidden areas. A logical copy captures only files visible to the file system and can miss deleted or hidden material. If full physical acquisition is impossible (hardware encryption, damaged media), the examiner must record the acquisition type actually achieved and its limitations.

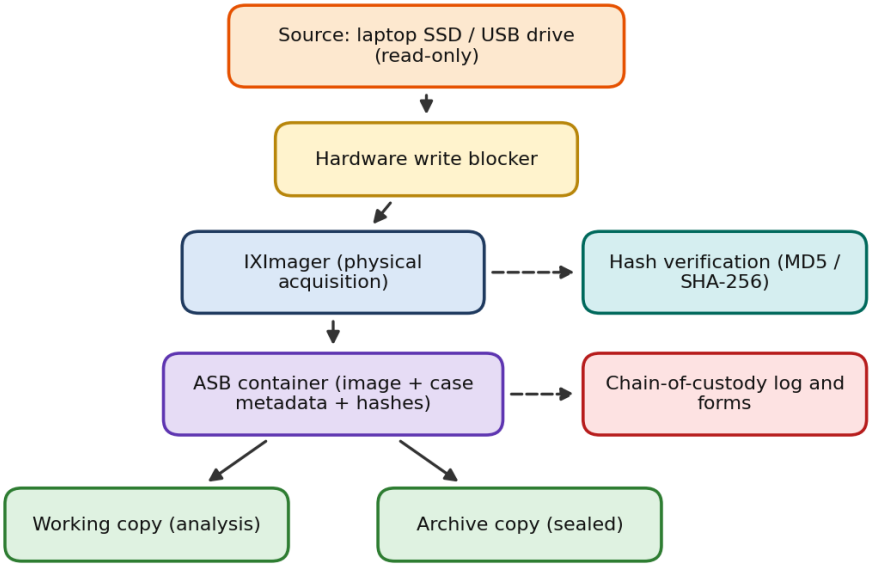
Analysis is performed on a **forensic image** because (a) the original stays unchanged and therefore reliable, (b) the image can be verified by hash, (c) a damaged working copy can be recreated, (d) several examiners can work independently, and (e) booting or mounting the original can change timestamps and write data.

Q3. Imaging workflow using IXImager

Step	Action	Record
1. Source identification	Identify the SSD (removed from the laptop) and the USB by model, serial, capacity and interface; confirm it is the right device before starting	Device details, photograph, examiner
2. Write protection	Connect through a hardware write blocker; confirm the device appears read-only	Blocker model and serial
3. Configure	Start IXImager; choose source and a separate destination with free space; choose ASB container, hash algorithms and enter case metadata	Case/item numbers, settings
4. Acquire	Run the physical acquisition; monitor read errors and bad sectors	Start/end time, errors
5. Verify	Let the tool compare the source hash with the image hash	Both hash values, result
6. Hash and secure	Log hash values; make a second copy; seal the original	Copy locations and hashes
7. Document	Save the acquisition report, sign it, attach to the case file and custody form	Report ID, signatures

The same workflow is repeated for the USB drive as a separate evidence item with its own container and hash values. For the BitLocker volume, the record states whether the image is of the decrypted (live) volume or of the encrypted volume together with the recovery key used.

Figure 4: Imaging and evidence preservation architecture



Q4. Hash values and integrity before and after acquisition

A cryptographic hash (for example SHA-256) produces a fixed-length fingerprint of data; if the same data is hashed again the value must match, and changing even one bit changes the value completely. A match supports, with extremely high confidence, that the compared data is unchanged. It does not show that the data was truthful or who created it. The report records algorithm, value, tool/version, date/time, object hashed and the verification result.

Stage	Record required	Purpose
Before acquisition (if feasible)	Source identifiers and any source hash the method supports	Baseline, or reason why none exists
During acquisition	Tool log, settings, read errors, start/end time	Makes the process reproducible
After acquisition	Image hash and verification result	Compare image with acquisition record
During transfer	Hash of the transferred copy and receiver's confirmation	Integrity while moving between people or places
Before analysis and reporting	Working-copy hash compared with the master image hash	Confirms the analysis copy equals the verified master

A mismatch means corruption, a faulty cable or blocker, or tampering; it requires re-imaging and a documented explanation. Using two algorithms (for example MD5 and SHA-256) is optional extra assurance.

Q5. Role of the ASB container

The case refers to an **ASB container** as the format in which acquired data is stored or managed with the IXImager workflow. Its exact features and supported metadata must be confirmed from the product documentation rather than assumed. In general, such a container keeps the image, its verification data and descriptive details together so that they stay consistent, but it does **not** replace hash verification or the chain-of-custody record. Information that should accompany the evidence:

- Case ID, evidence ID; source device make, model, serial, capacity.
- Acquisition date/time, examiner, location, method, tool name and version, settings, time zone and clock offset.
- Container/image format, segmentation or compression used.
- Hash algorithm and values, verification status, read errors or unreadable areas.
- Source condition, write-protection method, any live-response or shutdown decisions, encryption status.
- Custody history, storage location, access permissions and transfer records.

Q6. Data carving and when it is necessary

Data carving recovers files or fragments by scanning raw data for structural patterns such as file headers and footers (for example JPEG starts with FF D8 FF; PDF with %PDF; ZIP/DOCX/XLSX with 50 4B 03 04; Windows executables with MZ) without relying on intact file-system records. It is needed when file-system information is missing or damaged: after quick formatting, a corrupted partition table or MFT, deleted files whose directory entries are overwritten, or when examining unallocated and slack space.

Limits: carved files usually lose names, paths and timestamps; fragmented files may be incomplete or wrong; false positives occur. Carved items are stored separately, hashed, validated by checking file structure and content, and reported with the method, image offset and limitations.

Q7. Date and time problems and how to document them

- **Clock error:** photograph the laptop clock at seizure and compare with a trusted time source; record the offset.
- **Time zones and daylight saving:** record the configured zone (for example IST, UTC+05:30). NTFS stores times in UTC; FAT stores local time; convert everything to one reference.
- **Timestamp types:** identify whether a value is creation, modification, access, log or application time; copying and extraction change them differently.
- **Tampering or drift:** manual clock changes, restored backups and sync can distort times; compare with logs and journal entries.
- **Report practice:** keep original timestamps, show normalised times separately, state the time source, offset and assumptions, and mark uncertain entries instead of implying exact order.

Item to record	Value (hypothetical example)
Trusted time source	Synchronised reference clock / official time source
Laptop clock at seizure	Shown by photograph, e.g. 13 Oct 2026 11:23:40
Calculated offset	System time minus reference time, e.g. +2 min 10 s
Laptop time-zone setting	IST (UTC+05:30), daylight saving off
File-system time convention	NTFS (UTC)
Adjustment in timeline	State the correction applied; mark uncertain entries

Q8. Chain-of-custody table for the seized laptop

Date / time	Item	Released by (sign)	Received by (sign)	Purpose / location	Seal / condition
13 Oct 11:20	L-01 laptop (L-02 USB bagged separately)	Mr. T. Ganesan (collector)	Ms. Anitha V. (custodian)	Seizure and packaging, Network Operations Room	Bag A-201 sealed; live capture noted
13 Oct 12:40	L-01 laptop	Ms. Anitha V.	R. Arulkumar (examiner)	Transport to forensic lab	Seal intact, signed
13 Oct 13:05	L-01 laptop	R. Arulkumar	Lab evidence locker 2	Storage pending imaging	Seal intact, logged
14 Oct 09:30	L-01 laptop and SSD	Evidence locker 2	R. Arulkumar	Imaging with write blocker	Seal broken and re-sealed (new seal A-209), reason noted
14 Oct 16:45	L-01 original (SSD refitted)	R. Arulkumar	Lab evidence locker 2	Return after imaging	Hash verified; sealed
15 Oct 10:00	Image IM-01 (working copy)	R. Arulkumar	M. Farhan (analyst)	Examination of working copy	Hash re-verified, signed
15 Oct 10:05	Image IM-01 archive copy	R. Arulkumar	Evidence custodian	Sealed archive storage	Hash recorded, signed

Conclusion for Assignment 2(3)

Reliable imaging depends on correct source identification, write protection, a documented physical acquisition, hash verification, secure storage and an unbroken custody record. IXImager and the ASB container must be used according to their documented capabilities, and every error or limitation must be disclosed.

Assignment 2(4): Recovery of Deleted Evidence from Volume Shadow Copies

Unique scenario selected: *Alteration and deletion of Scholarship_Disbursal_Oct2026.xlsx on the Accounts Section workstation ACC-PC01, with earlier versions recovered from two Volume Shadow Copies.*

Assumptions and case-file identifiers (hypothetical)

Field	Value
Case number	GEC-CDF-2026-A4
Subject	Mr. Balaji N., Junior Assistant, Accounts Section (suspect; no conclusion is assumed)
Workstation	ACC-PC01 – desktop PC, Accounts Section, Admin Block; Windows 10 Pro; data volume D:
File in question	D:\Accounts\Scholarship\Scholarship_Disbursal_Oct2026.xlsx (original version V1, SHA-256 a41f...09c2, hypothetical)
Removable device seen	Kingston DataTraveler 32 GB (serial from USBSTOR)
Snapshots	S1 created 02 Oct 2026 22:00 (scheduled backup); S2 created 09 Oct 2026 22:00 (scheduled backup)
Reported by	Internal auditor, 12 Oct 2026, 09:20 IST, after bank-account numbers of nine beneficiaries did not match the sanction list
Material available	Verified forensic image of the system and data volumes; image hash recorded in the acquisition log; authorisation Ref. GEC/PRIN/2026/125

The timeline and values are illustrative. Tool steps describe the intended workflow, not operations actually performed.

Q1. Selecting and categorising evidence before detailed analysis

1. **Define the questions:** what changed, by which session, when, was it copied, how was it removed, and what can be recovered?
2. **Inventory** all volumes, user profiles and snapshots in the image.
3. **Eliminate noise:** remove known operating-system and application files with a hash set; do not delete anything, only filter the view.
4. **Filter** by user profile, date range (29 Sep to 13 Oct), file type and keywords such as "Scholarship", "Disbursal".
5. **Categorise** remaining items by relevance and reliability; record why items were excluded so the decision can be reviewed.
6. **Prioritise** high-value items: the workbook and its versions, Office recent-file data, browser databases, event logs, USB artifacts and snapshots.
7. **Minimise** exposure of unrelated personal or financial data that is outside the authorised scope.

Q2. Classification scheme

Category	Definition	Examples in this case	Handling
User-created files	Files produced or edited by a person	Scholarship_Disbursal_Oct2026.xlsx, drafts, AutoRecover copies	Preserve, hash, compare content and metadata
System / application files	Created automatically by Windows or programs	Event logs, registry hives, Prefetch, Office recent-file list, jump lists	Use as activity evidence; filter known-good files by hash set
Potentially suspicious files	Unusual by name, place, time or behaviour	A disk-cleaning utility and its Prefetch entry, renamed executables	Examine closely; check install and execution traces; do not assume malicious from name alone
Deleted / recovered artifacts	No longer visible in the live volume but retrieved	Workbook versions V1 and V2 from snapshots, $I/$R Recycle Bin records, earlier browser history	Record source (snapshot ID or unallocated space), hash, validate, label as recovered
Supporting / contextual evidence	Adds background; does not prove the action alone	Login schedule, door-access records, audit complaint, sanction list	Use to explain opportunity and to corroborate
Out-of-scope material	Unrelated personal or business data	Personal photos, unrelated student records	Do not collect or include without a legitimate reason

Q3. How Event Analysis helps reconstruct a sequence of activities

Event Analysis assembles time-stamped records from many sources (Windows event logs, file-system timestamps, registry, browser history, Office artifacts and snapshot contents) into one chronological view. It answers what happened first, what followed, and which user session each event belongs to. In this case a chain such as *logon, workbook opened, edited and saved, copy to USB, cleaner tool run, Recycle Bin emptied, history cleared* becomes a connected story. Original timestamps are kept beside the normalised UTC/IST value, and gaps, conflicting clocks, duplicates and cleared logs (Event 1102) are flagged. Close timing supports a hypothesis but does not prove causation.

Source	Event or artifact	Value
Security log	4624 / 4634 logon and logoff	Defines user sessions
Office recent files, jump lists	Workbook opened; time and path	Shows the file was opened and in which profile
Recycle Bin ($I and $R)	Original path and deletion time	Shows when the file was deleted
Prefetch	Execution of a cleaner tool	Shows tool use and time
Browser databases	Visits, downloads, cleared entries	Web activity around the deletion
USBSTOR, LNK	Device connection and file copy trace	Possible data movement

Q4. Cloud Analysis and Lead Analysis

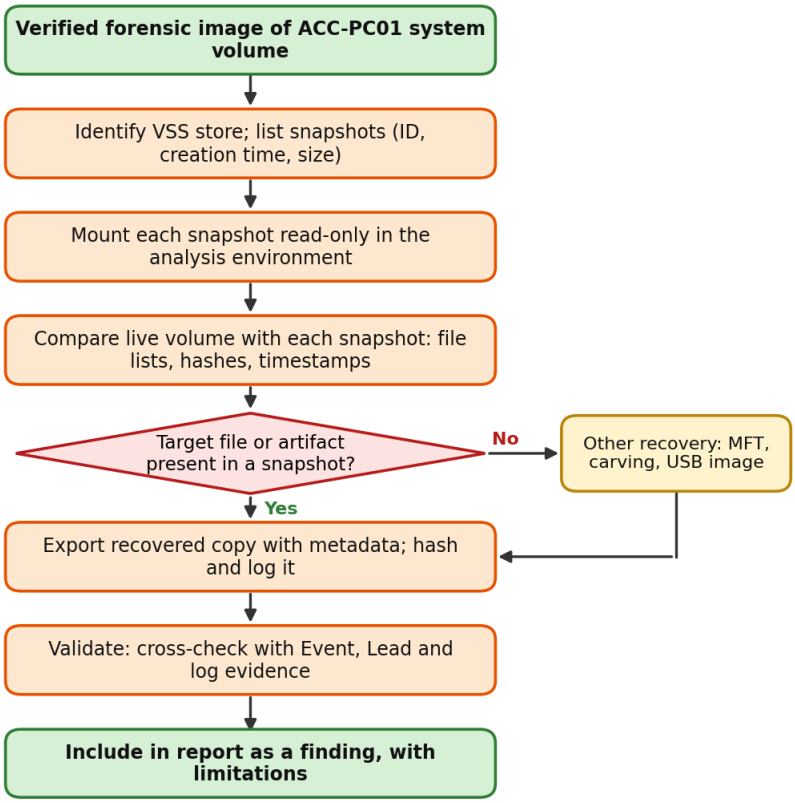
Technique	Purpose	Questions it can help answer
Cloud Analysis	Examines evidence of cloud services used from the system (storage, webmail, sync clients) and their local traces; extends to provider records only where lawfully available	Was the workbook uploaded or synchronised to a cloud account? Which account and when? Is another copy or version history held in the cloud? Which cloud apps were installed?
Lead Analysis	Uses names, e-mail addresses, URLs, phone numbers and keywords found in the data to point to further evidence sources and testable questions	Who did the user communicate with? Which addresses and sites recur? Is there another device, account or person to examine? Which other copies of the same hash exist?

Leads are directions for investigation, not conclusions, until corroborated.

Q5. How Volume Shadow Copy analysis recovers or compares earlier versions

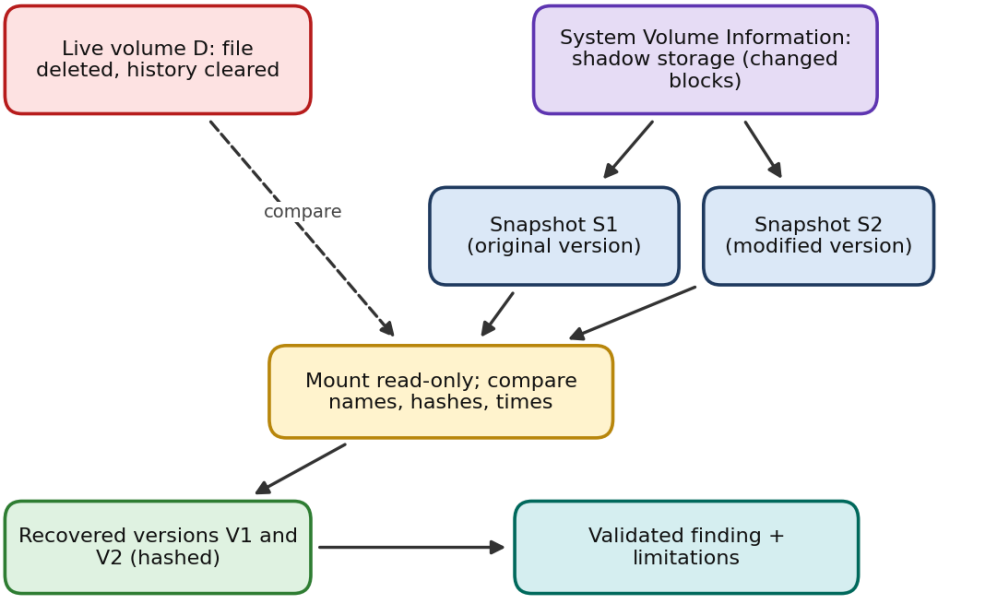
The Volume Shadow Copy Service keeps point-in-time views of a volume. Instead of copying the whole volume it preserves the old content of blocks that are about to change in a shadow storage area, so each snapshot presents the volume as it was at creation. Investigators can (a) recover a file deleted from the live volume if it existed at snapshot time, (b) compare a file across snapshots to see what changed, and (c) open earlier browser databases, registry hives or logs that were later cleared. The usual process is to list snapshots, mount them read-only, compare against the live volume by name, hash and timestamps, then export items with their source recorded.

Assignment 2(4): Volume Shadow Copy recovery workflow



Flowchart 4: Volume Shadow Copy recovery workflow

Figure 5: How earlier versions survive in Volume Shadow Copies

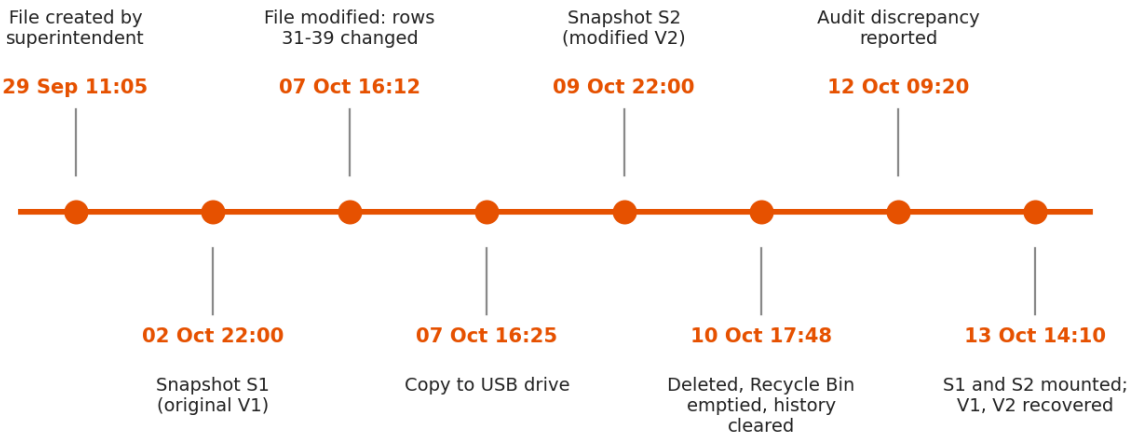


Comparison	Method	Question answered
File presence	List files in each snapshot and the live volume	Which files exist in one but not the other?
File content	Compare hashes of the same path	Was content changed between two points in time?
Metadata	Compare MACB times and size	When and by how much was it changed?
Browser databases	Open History/Downloads from each snapshot	Which entries were removed later?

Comparison	Method	Question answered
Event logs and hives	Compare Security log and NTUSER.DAT/SYSTEM	Were logs cleared? Which devices or programs were used earlier?

Q6. Hypothetical timeline: create, modify, copy, delete, recover

Figure 6: Hypothetical timeline for Scholarship_Disbursal_Oct2026.xlsx (IST)



Date / time (IST)	Event	Source of evidence	Comment
29 Sep 11:05:42	Workbook created in D:\Accounts\Scholarship by the section superintendent	NTFS created time; Office author metadata	Original version V1
02 Oct 22:00:00	Snapshot S1 created (scheduled)	VSS snapshot record	Contains V1
07 Oct 16:12:30	Workbook modified: nine rows (31 to 39) changed	Modified time; Office metadata (last modified by); AutoRecover	Changed version V2; hash differs from V1
07 Oct 16:25:10	File copied to USB drive	USBSTOR; LNK file; jump list	Possible data movement
09 Oct 22:00:00	Snapshot S2 created (scheduled)	VSS snapshot record	Contains V2
10 Oct 17:48:20	File deleted; Recycle Bin emptied; cleaner tool run; browser history cleared (146 entries)	MFT entry marked free; $I/$R; Prefetch; browser database times	Anti-forensic act or routine clean-up
12 Oct 09:20	Auditor reports mismatch	Complaint register	Discovery time
13 Oct 14:10-14:40	S1 and S2 mounted read-only; V1 and V2 exported, hashed and compared	Examiner log	Differences limited to rows 31-39

Because S2 was created *after* the modification and *before* the deletion, both versions can be recovered from snapshots. Snapshot time is not the file's creation or edit time; it only shows that the version existed by then.

Q7. Validating a recovered artifact before it becomes a finding

#	Validation check	Done
1	Confirm the source image hash matches the acquisition record	[ ]
2	Record snapshot ID, volume, creation time and the read-only mount method	[ ]
3	Hash the recovered file and log the value	[ ]
4	Confirm the file opens and its content is complete and readable	[ ]
5	Compare metadata (author, created, modified, last modified by) with other artifacts	[ ]
6	Corroborate with at least one independent source (event log, USB record, Recycle Bin, browser data)	[ ]
7	Repeat the recovery with a second tool or a second run to check reproducibility	[ ]
8	Check time-zone and clock offsets on every timestamp used	[ ]
9	Write down limitations and alternative explanations	[ ]
10	Peer review by a second examiner before inclusion in the report	[ ]

Q8. Sources of error and alternative explanations

- **Automatic deletion:** a clean-up policy, disk clean-up or sync client could have removed the file rather than the suspect.
- **Snapshot limits:** a snapshot shows state only at creation time; changes between snapshots are invisible; old snapshots may be deleted when space is full.
- **Timestamp reliability:** clock changes, time-zone differences (UTC versus IST), copy/restore operations and tools that reset times can mislead; snapshot time is not edit time.
- **Shared account or remote access:** someone else may have used the account, or a program may have acted automatically.
- **Incomplete browser evidence:** private browsing, profile sync or cleared history means missing entries do not prove they never existed.
- **Recovery artefacts:** partly overwritten or fragmented recovered files may contain errors; parsing bugs and wrong path associations can cause misinterpretation.
- **Presence is not intent:** a deleted file in a snapshot does not prove who deleted it.

Conclusion for Assignment 2(4)

Volume Shadow Copies give valuable historical context, but each recovered artifact must keep its source snapshot and provenance. A defensible report compares versions, correlates several sources, validates the extraction, and separates observed facts from inferred activity.

Assignment 2(5): Multi-Source Investigation of a Suspected Cyber Incident

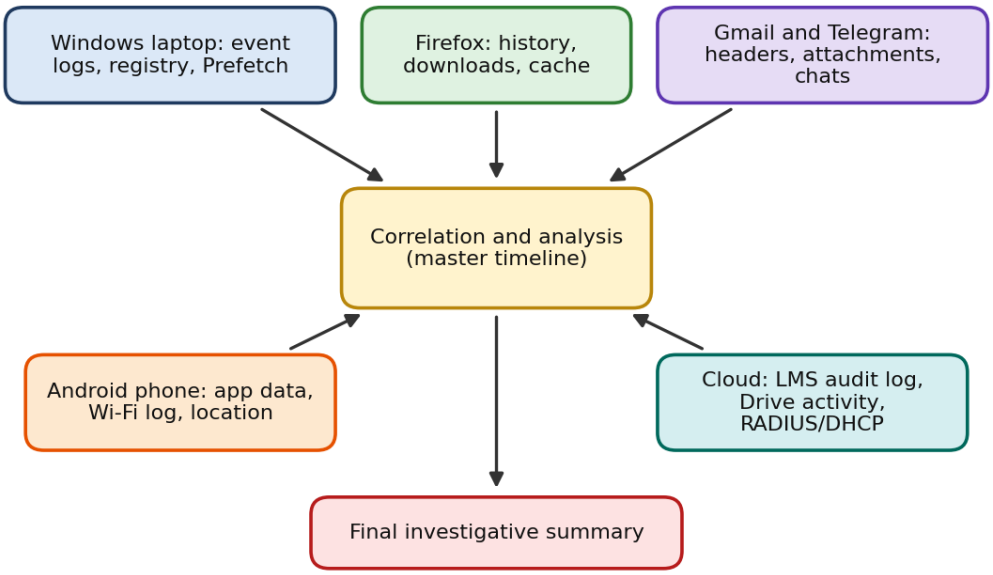
Unique scenario selected: *Unauthorized faculty-level login to the college learning-management system (GEC-LMS) and alteration of internal-assessment marks, with the student account arvind.p under suspicion.*

Assumptions and case-file identifiers (hypothetical)

Field	Value
Case number	GEC-CDF-2026-A5
Service	GEC-LMS (Moodle-based), hosted by the college; the faculty account faculty.cse12 shows a login and grade edits that the faculty member denies
Student account	arvind.p (final-year B.E. student, hypothetical); suspicion arises because the source IP lease belongs to his laptop
Laptop	Lenovo IdeaPad Slim 3, Windows 11 Home, Firefox as main browser
Phone	Android 13 phone (IMEI recorded at seizure)
Accounts	Gmail (student's personal), Telegram, Google Drive; college RADIUS/DHCP logs for the campus Wi-Fi
Incident window	Sunday 04 Oct 2026, evening; LMS audit log shows faculty.cse12 login 21:17:40 and logout 21:26:12; six grade items edited
Reported by	Faculty member and the Head of Department, Mon 05 Oct 2026
Authorisation	Ref. GEC/PRIN/2026/129: laptop, phone, browser data, e-mail and messaging records, provider and cloud logs, as far as consent or legal process permits

All artifacts, IDs and times are hypothetical. Evidence is examined only under authorisation; tool steps describe intended workflow.

Figure 7: Multi-source evidence map for the GEC-LMS incident



Multi-source evidence map (deliverable: evidence map)

Q1. Potential evidence from Windows and other operating systems

System	Evidence types
Windows	Event logs (4624/4625 logons, process creation, USB, Wi-Fi), registry (UserAssist, RecentDocs, ShimCache, network profiles), Prefetch, jump lists and LNK files, shellbags, Recycle Bin, hibernation and page files, scheduled tasks, installed tools, Volume Shadow Copies, browser databases
Linux / macOS (if used)	Shell history, authentication logs, cron jobs, package history, application logs, home directories; unified logs and plist files on macOS
Mobile OS (Android / iOS)	App data, call and SMS logs, location history, Wi-Fi connection records, installed apps, system logs, app-specific databases
Across all systems	Record OS version, device identity, configured time zone, account identifiers, acquisition method and limitations. An OS artifact usually identifies a device, account or process, not the person at the keyboard

Q2. Browser artifacts and why they need careful interpretation

Browsers may hold history, URLs, search terms, downloads, cookies, cache, bookmarks, autofill, saved logins, session data, extensions and profile metadata. In this case they can show a visit to the LMS login page, the time and frequency, a downloaded file and which account was signed in. They must be interpreted carefully because history can be cleared or synchronised from another device, private browsing leaves fewer traces, a URL in history does not prove that a login succeeded or that the account holder initiated it, cache may come from auto-loaded content, timestamps use browser-specific formats and time zones, and scripts or extensions can create entries without deliberate action.

Q3. E-mail and messaging evidence and how it correlates

- **E-mail:** sender and recipient fields, Message-ID, Date and Received chain (read bottom to top), originating IP, SPF/DKIM/DMARC results, body, attachments and their hashes, folders, drafts, deleted items and account login/audit records.
- **Messaging:** account identifiers, message timestamps, participants, shared files and links, delivery/read state, device and session information, local app databases, where lawfully available.
- **Preservation:** keep original exports or native records with acquisition method and metadata; screenshots are exhibits, not substitutes.
- **Correlation:** match a message containing the LMS URL with the browser visit time; match attachment hashes with local files, Drive objects and downloads; match header IPs with server logs; match account identifiers across services; consider forwarding, spoofing, shared accounts, synchronisation and clock differences.

Q4. Forensic considerations for the mobile phone and its challenges

- Begin with authorisation, device identification (make, model, IMEI, OS version), condition, lock state and battery; photograph the device.
- Decide network isolation and power state (airplane mode or Faraday bag) to prevent remote wipe, while keeping the device powered.
- Consider passcode, full-disk or file-based encryption, cloud backups, app-specific storage, SIM/eSIM identifiers, and the effect of the extraction method on data.
- Use a validated, documented method and record whether the result is a **logical, file-system, backup-based or physical** extraction; these do not recover identical data.
- Do not claim access to deleted or encrypted content unless the extraction actually supports it; acquire cloud-linked data only within the authority given.

Challenges: screen locks and encryption, frequent OS updates that break tool support, protected app storage, many device models, auto-deleting messages, remote wipe, and cloud-synchronised data that exists off the device.

Q5. Internet and cloud-based evidence sources

Source	Data available	How obtained	Constraint
GEC-LMS server	Web, authentication and grade-edit audit logs, session IDs, IP, user agent	Direct from college administrators	Log retention period
Campus network devices	DHCP leases, RADIUS/Wi-Fi authentication, firewall logs (IP to MAC and time)	Network team	Time synchronisation needed
Internet service provider	IP-to-subscriber records for external access	Legal process or police request	Retention, jurisdiction
E-mail provider	Message data, login and device logs	Account consent or legal process	Provider policy, data location
Cloud storage (Google Drive)	File version history, sharing and access logs, device list	Account access with consent or legal process	Encryption, retention
Messaging provider	Account and activity metadata	Legal process	End-to-end encryption limits content

Provider data usually needs legal process, retention can be short, and IPs may be shared, dynamic or masked by VPNs, NAT and proxies, so early preservation requests are important.

Q6. Cross-source correlation table (hypothetical artifacts)

#	Source	Artifact	Time (IST, 04 Oct)	Correlates with / caution
1	Google Drive (cloud)	Activity log: file marks_edit_plan.xlsx last modified from the student's account	20:49:10	Intent/knowledge before the event; sync client or another device could also edit
2	Gmail (e-mail)	Sent message to a friend containing the LMS gradebook URL	20:38:25	Header IP and Message-ID; sending does not prove the sender then logged in
3	Telegram (messaging)	Chat message "will check after 9" with a screenshot attachment (hash recorded)	20:55:02	Attachment hash against local file; message timestamps may be device-local
4	Android phone (mobile)	Wi-Fi association with SSID GEC-Campus; mobile browser tab on the LMS URL	21:05:30	Matches DHCP lease; app timestamps may be delayed or synchronised
5	Windows laptop	Event 4624 for the student's local account; Firefox Prefetch entry	21:12:48	Session active before the LMS login; shows account, not person
6	Firefox profile (browser)	History: LMS login page visit; no download of the grade report	21:16:55	A history entry does not prove successful authentication
7	LMS server audit log (cloud/provider)	Login as faculty.cse12 from 172.20.15.64; six grade items edited; logout	21:17:40-21:26:12	Check time zone and whether the log identifies a device reliably
8	Campus DHCP/RADIUS	Lease binds 172.20.15.64 to the laptop's MAC address	20:45-22:00	Ties the LMS source IP to the device; a MAC can be spoofed

Q7. Major challenges when evidence is spread across devices and cloud services

1. **Time zones and clock drift:** devices and services record different times; keep original values and document the normalisation.
2. **Encryption and locked devices:** content may be unreachable without keys or an authorised extraction method.
3. **Cloud-provider dependence:** records may be unavailable, retained briefly, or need formal legal process and cooperation.
4. **Data volume and heterogeneity:** formats, schemas and app versions differ, making collection and correlation complex.
5. **Attribution:** shared devices, shared or stolen credentials, VPNs, proxies and dynamic IPs complicate identification of a human actor.
6. **Integrity and provenance:** exports, screenshots and conversions can lose context or metadata unless documented.
7. **Deleted or transient evidence:** logs, caches, messages and volatile data may be overwritten or expire.

8 **Privacy and scope:** devices hold unrelated personal data; collection must be minimised.

9 **Synchronisation and duplication:** one event can create artifacts on several devices, and a synchronised artifact may not reflect direct interaction on each.

10 **Tool limitations:** parsers may not support every app version or may misread timestamps and proprietary databases.

Q8. Final investigation conclusion

Category	Findings (based on the hypothetical artifacts above)
Direct evidence	The LMS audit log (provenance and time zone to be confirmed) records a login as faculty.cse12 and six grade edits between 21:17:40 and 21:26:12 from 172.20.15.64. The campus DHCP/RADIUS record binds that IP to the student's laptop during the same period.
Corroborating evidence	Local session on the laptop was active; Firefox history shows a visit to the LMS login page just before; the Drive activity, e-mail and Telegram message before the event are consistent with planning; the phone's Wi-Fi record and mobile browser tab fit the same period.
Unresolved questions	Who was physically operating the laptop; how the faculty credentials were obtained (phishing, shared password, shoulder surfing); whether the grades were altered by the same session or by another; whether data left the campus; whether a MAC address or IP was spoofed; and whether another device used the phone's connection.

Summary: the combined evidence *supports* the conclusion that the laptop associated with the arvind.p account was on the campus network and used to reach the LMS at the time of the unauthorised faculty login. It does not by itself prove that the student personally performed the act. Further authorised steps (examining the credential source, door-access or CCTV records, an interview with the student and the faculty member) are recommended before any disciplinary decision, and these limits are stated in the report.

Conclusion for Assignment 2(5)

A multi-source investigation is strongest when independent systems converge on the same event and the provenance of every artifact is preserved. Careful time normalisation, validation, privacy protection and clearly stated limitations are essential to a fair and defensible conclusion.

Appendix A: Glossary of Terms

Term	Meaning
Digital evidence	Information stored or transmitted in digital form that may be used in an investigation or proceeding
Chain of custody	Documented record of who handled the evidence, when, where and why
Forensic image	Bit-for-bit copy of a storage device or volume
Write blocker	Hardware or software that allows reading from a device while preventing writes
Hash value	Fixed-length fingerprint computed from data, used to verify integrity
Order of volatility	Principle of collecting the most short-lived data first, such as memory before disk
Metadata / MACB	Data about data; Modified, Accessed, Changed (record) and Born (created) timestamps
Slack / unallocated space	Unused space at the end of a file's last cluster / disk space not assigned to any file; both may hold older data
Data carving	Recovering files from raw data using signatures instead of file-system records
Volume Shadow Copy	Windows feature that keeps point-in-time snapshots of a volume
Artifact	Trace of user or system activity left in files, logs, registry or databases
ASB container	Forensic evidence container used with the course tools to keep an image and related data together (confirm details in product documentation)
ILookIX / IXImager	Forensic analysis and imaging tools referred to in the CR23532 syllabus
Attribution	Linking an activity to a specific person, not only to an account or device
Time-zone offset	Difference between a local time and UTC (IST is UTC+05:30)

Appendix B: Standards, Guidelines and Laws Referred To

- NIST SP 800-86: Guide to Integrating Forensic Techniques into Incident Response.
- ISO/IEC 27037: Guidelines for identification, collection, acquisition and preservation of digital evidence.
- ACPO Good Practice Guide for Digital Evidence: original data must not be changed; keep an audit trail.
- Information Technology Act, 2000 (India): offences such as unauthorised access and computer-related offences (for example Sections 43 and 66).
- Bharatiya Sakshya Adhiniyam, 2023: admissibility of electronic records, including the certificate requirement.

Overall Conclusion

The five case studies follow the life cycle of a digital investigation: lawful planning and collection, preservation and imaging, examination of file-system artifacts and recovery of deleted data, and analysis of evidence spread over several devices and services, ending with reporting. The main lessons are that evidence must be preserved and verified before it is analysed; that every finding should rest on more than one independent artifact; that timestamps need careful handling; and that artifacts identify accounts and devices, so claims about a person must be cautious and state their limits openly.